ECAS Summer Course

Fundamentals of Quantile Regression

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An Outline

- Beyond Average Treatment Effects
- Computation of Quantile Regressions
- Some Equivariance Properties
- Caution: Quantiles Crossing!
- Simulation and the Random Coefficient Interpretation
- Gini, Lorenz and the Decomposition of Inequality

Beyond Average Treatment Effects

Lehmann (1974) proposed the following general model of treatment response for the two-sample randomized treatment-control model:

> "Suppose the treatment adds the amount $\Delta(x)$ when the response of the untreated subject would be x. Then the distribution G of the treatment responses is that of the random variable $X + \Delta(X)$ where X is distributed according to F."

Lehmann QTE as a QQ-Plot

Doksum (1974) defined $\Delta(x)$ as the "horizontal distance" between F and G at x, *i.e.*

$$F(x) = G(x + \Delta(x)).$$

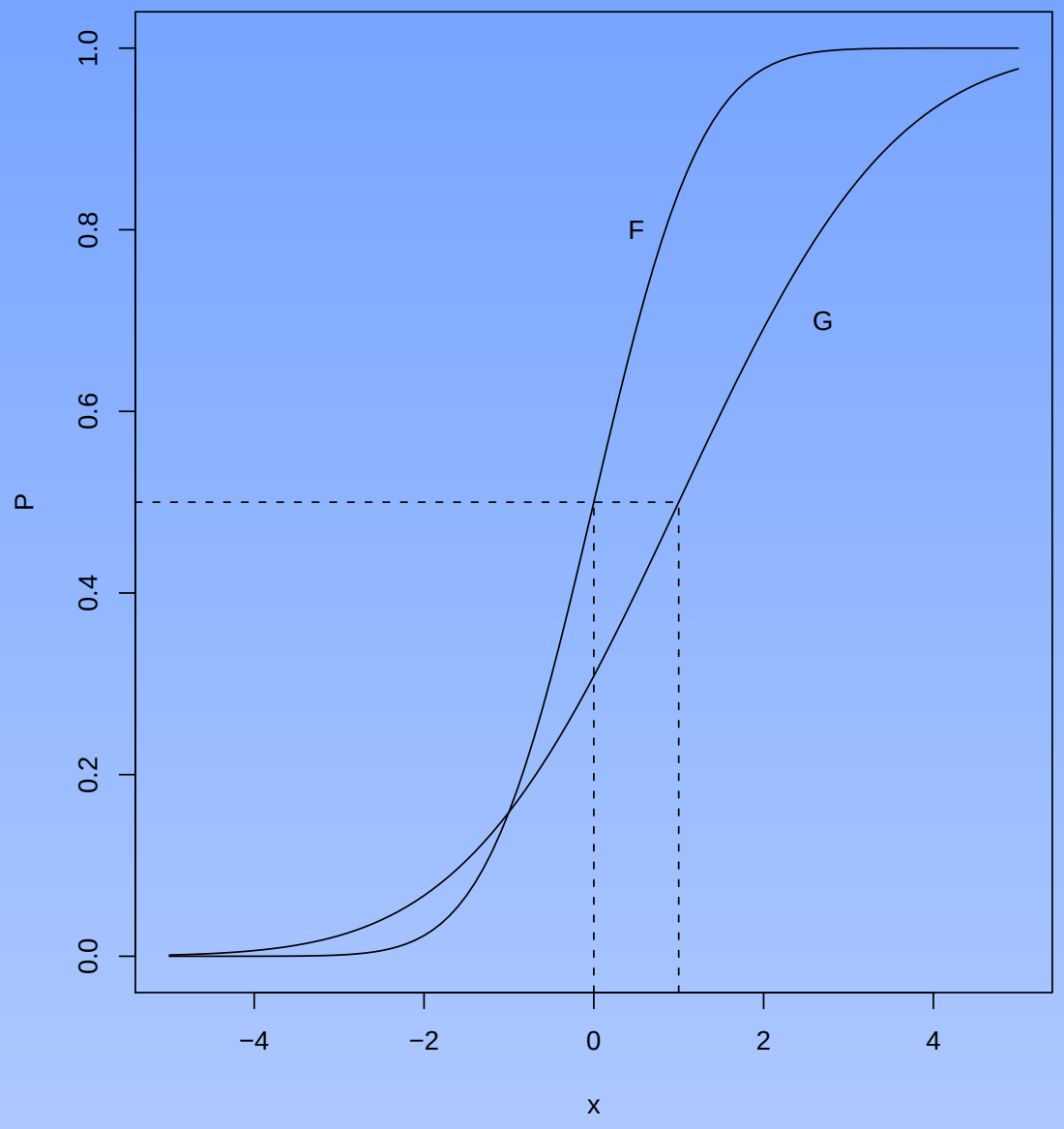
Then $\Delta(x)$ is uniquely defined as

$$\Delta(x) = G^{-1}(F(x)) - x.$$

This is essentially the conventional QQ-plot. Changing variables so $\tau = F(x)$ we have the quantile treatment effect (QTE):

$$\delta(\tau) = \Delta(F^{-1}(\tau)) = G^{-1}(\tau) - F^{-1}(\tau).$$

Lehmann-Doksum QTE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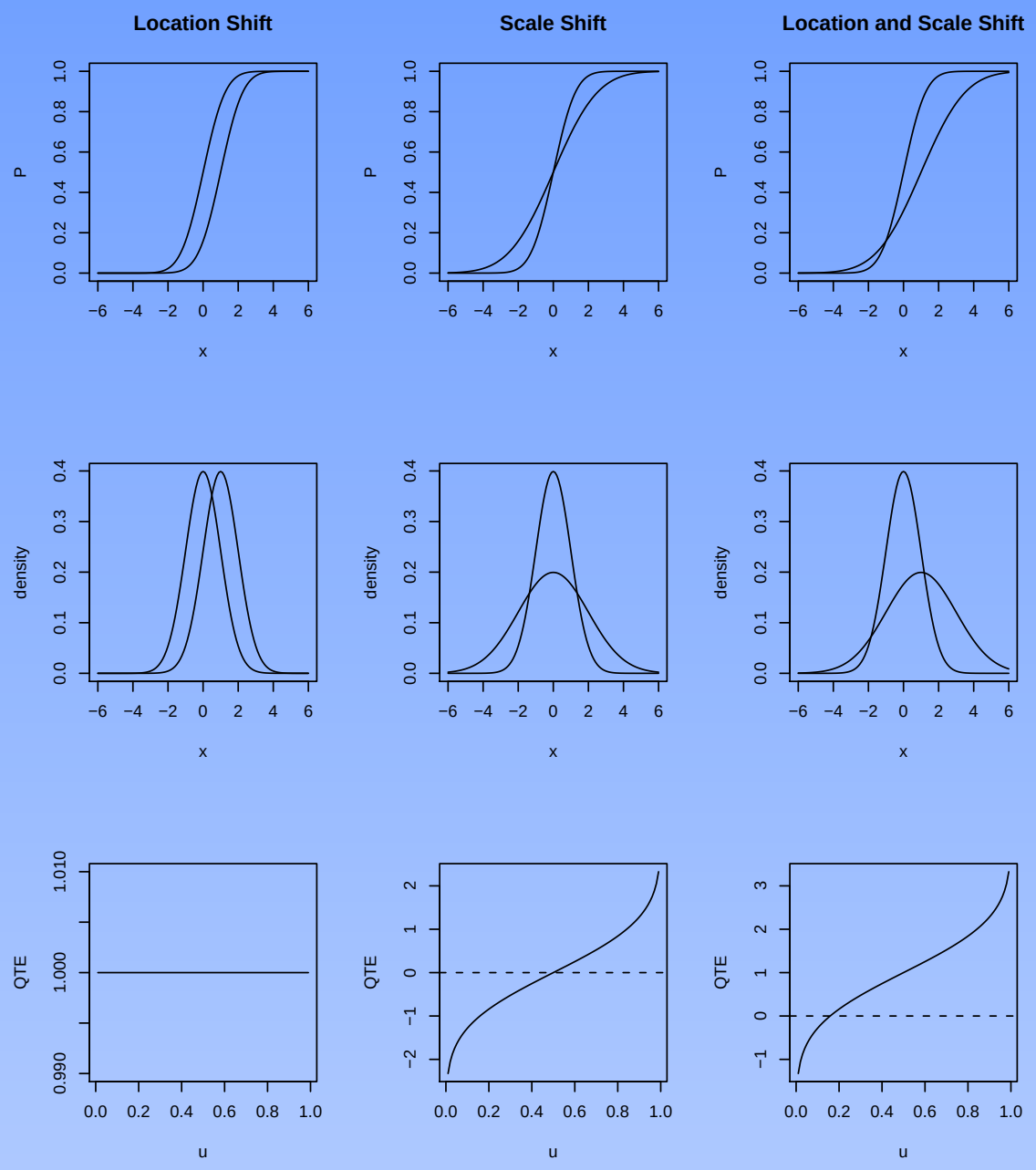
Location Shift
Scale Shift
Location and Scale Shift
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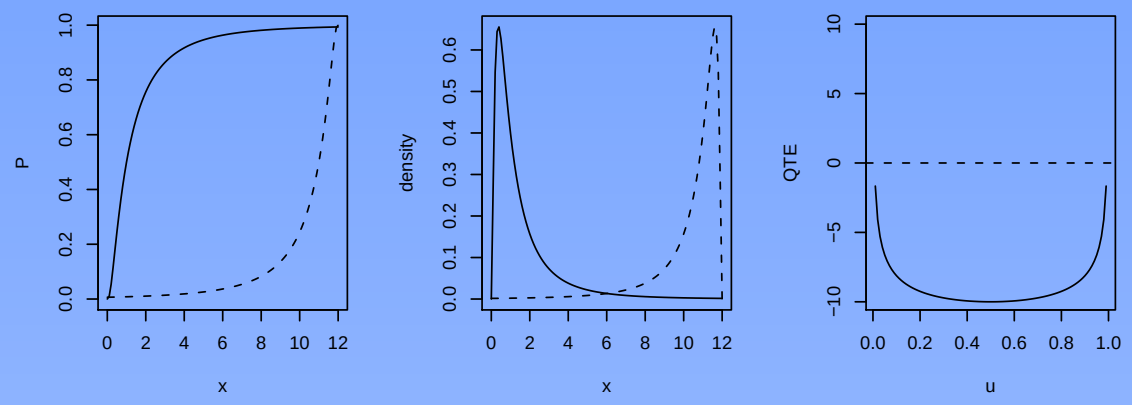
An Asymmetric Example



Treatment shifts the distribution from right skewed to left skewed making the QTE U-shaped. Control: $X \sim$ log-normal
Treatment: $Y = 12 - X$

The Erotic is Unidentified

The Lehmann QTE characterizes the difference in the marginal distributions, F and G, but it cannot reveal anything about the joint distribution, H. The copula function, Schweizer and Wolf (1981), Genest and McKay, (1986),

$$\varphi(u, v) = H(F^{-1}(u), G^{-1}(v)),$$

is not identified. Lehmann's formulation $assumes$ that the treatment leaves the ranks of subjects invariant. If a subject was going to be the median control subject, then he will also be the median treatment subject.

QTE via Quantile Regression

The Lehmann QTE is naturally estimable by

$$\hat{\delta}(\tau) = \hat{G}_n^{-1}(\tau) - \hat{F}_m^{-1}(\tau)$$

where $\hat{G}_n$ and $\hat{F}_m$ denote the empirical distribution functions of the treatment and control observations, Consider the quantile regression model

$$Q_{Y_i}(\tau|D_i) = \alpha(\tau) + \delta(\tau)D_i$$

where D_i denotes the treatment indicator, and $Y_i = h(T_i)$, $e.g.$ $Y_i = \log T_i$, which can be estimated by solving,

$$\min \sum_{i=1}^{n} \rho_\tau(y_i - \alpha - \delta D_i)$$

Computation of Quantile Regression

Primal Formulation as a Linear Program

$$\min\{\tau 1^\top u + (1-\tau)1^\top v | y = Xb + u - v, (b,u,v) \in \mathbb{R}^p \times \mathbb{R}^{2n}_+\}$$

Dual Formulation as a Linear Program

$$\max\{y'd | X^\top d = (1-\tau)X^\top 1, d \in [0,1]^n\}$$

Primal solutions are characterized by an exact fit to p observations. $b(h) = X(h)^{-1}y(h)$ where h indexes p element subsets of $\{1, 2, \cdots, n\}$.

Dual solutions have $0 < d_i < 1$ for $i \in h$ and $d_i = I(y_i < x_i^\top b(h))$ for $i \notin h$.

Optimality and the Subgradient Condition

Optimality of a basic solution $b(h) = X(h)^{-1}y(h)$ requires that

$$\nabla_v R(b) = \lim_{t \to 0} \frac{R(b + tv) - R(b)}{t} \geq 0$$

For convex, polyhedral functions like R this requires checking $2p$ directions corresponding to the choice of dropping one of the current elements of h and traveling in either the positive or negative direction. Optimality occurs when none of these directions succeeds in decreasing the objective function. In the language of convex analysis when zero is an element of the subgradient, $0 \in \partial R(b(h))$. For $\psi_\tau(u) = \tau - I(u < 0)$, essentially,

$$\sum_{i=1}^{n} \psi_\tau(y_i - x_i^\top b(h))x_i \equiv 0$$

The Gaussian Hare vs the Laplacian Tortoise

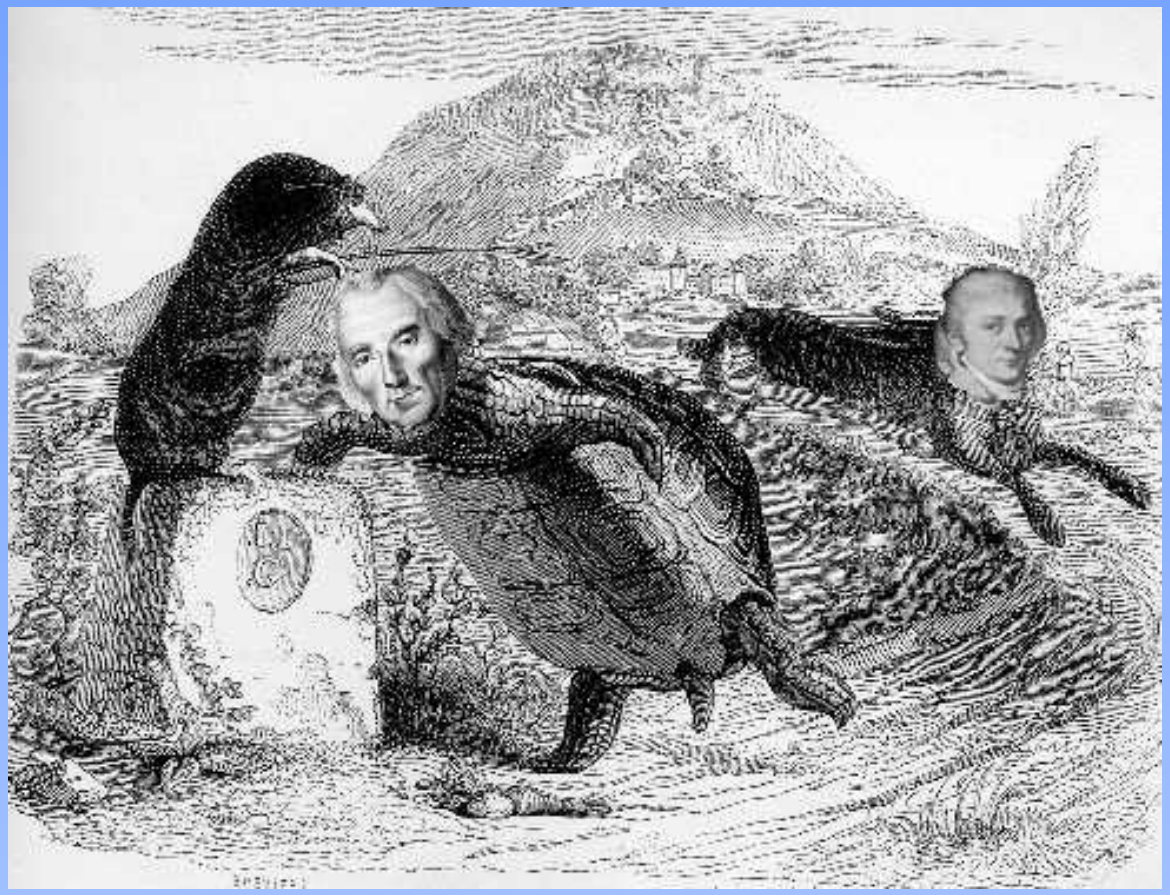


Figure 1: Retouched 18th century woodblock photo-print

Boscovich's Problem

Boscovich (1757) proposed estimating the ellipticity of the earth by solving:

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

subject to:

$$\sum_{i=1}^{n} (y_i - \alpha - x_i'\beta) = 0,$$

The constraint allowed the problem to be reduced to the weighted median problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i'\beta|$$

where $\tilde{y}$ and $\tilde{x}$ denote deviations from their means.

Laplace's *Methode de Situation*

Laplace showed that Boscovich's problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i' \beta|$$

could be solved by ordering the candidate slopes $\{\tilde{y}_i / \tilde{x}_i\}$ and finding the weighted median using weights $w_i = |\tilde{x}_i|$, i.e. finding the smallest j such that,

$$\sum_{i=1}^{j} w_{(i)} > \frac{1}{2} \sum_{i=1}^{n} w_{(i)}.$$

Edgeworth's Dual Plot and the Plural Median

F.Y. Edgeworth (1888) suggested solving

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

without Boscovich's constraint. His computational strategy introduced the "dual plot". Each pair of observations defined a point in parameter space, and lines in this space represent exact fits to one of the sample observations. Edgeworth suggested that a solution could be found by following a path of steapest descent through the dual plot. The multivariate version of this required "some power of hypergeometrical conception."

Dantzig's Simplex Method and the Cauchy Algorithm

Dantzig's (1948) simplex method for linear programming provided a general method for Edgeworth's plural median problem. Points in the general p dimensional setting corresponded to basic solutions, exact fits of p observations, and lines corresponded to solutions with $p-1$ zero residuals.

Traversing the dual plot by steapest descent is the simplex method.

Barrodale and Roberts (1978) algorithm implemented a particularly efficient version that can be interpreted as a form of the Cauchy algorithm: steapest descent at each step with one dimensional line search in the descent direction.

Frisch's Interior Point Radar

In the early 1950's Ragnar Frisch suggest that logarithmic barrier methods could be used for linear programming problems:

> My method is altogether different than simplex. In this method we work systematically from the interior of the admissible region and employ a logarithmic potential as a guide – a sort of radar – in order to avoid crossing the boundary.

This method anticipated modern "interior point" methods for linear programming. Particularly efficient for problems with sparse, design matrices, that is, X matrices that are mostly zeroes.

Interior versus Exterior Point Methods

Problem: Find the most northeast point in a circumscribed polygon. Formally:

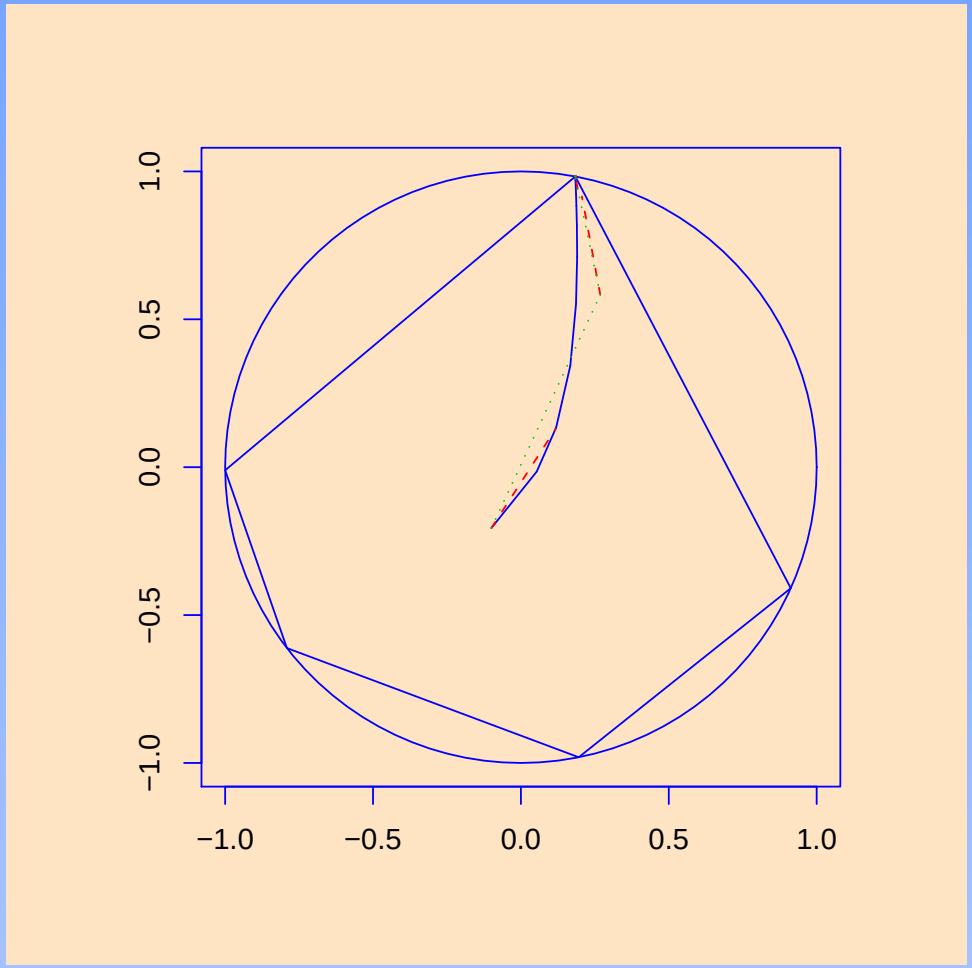
$$\max\{e'u | A'd = u,\ e'd = 1,\ d \geq 0\}$$

were e is vector of ones, and A has rows representing the n vertices. Eliminating u we can reformulate the problem as:

$$\max\{c'x | e'x = 1,\ \ x \geq 0\},$$

where $c = Ae$. Simplex is very simple here: we start at a random vertex and go around the edge of polygon until we find the optimum.

A Simple LP Problem



Interior versus Exterior Point Methods

In the interior point method we replace

$$\max\{c'x | e'x = 1, \;\; x \geq 0\},$$

with the log barrier reformulation

$$\max\{c'x + \mu \sum_{i=1}^{n} \log x_i | e'x = 1\}$$

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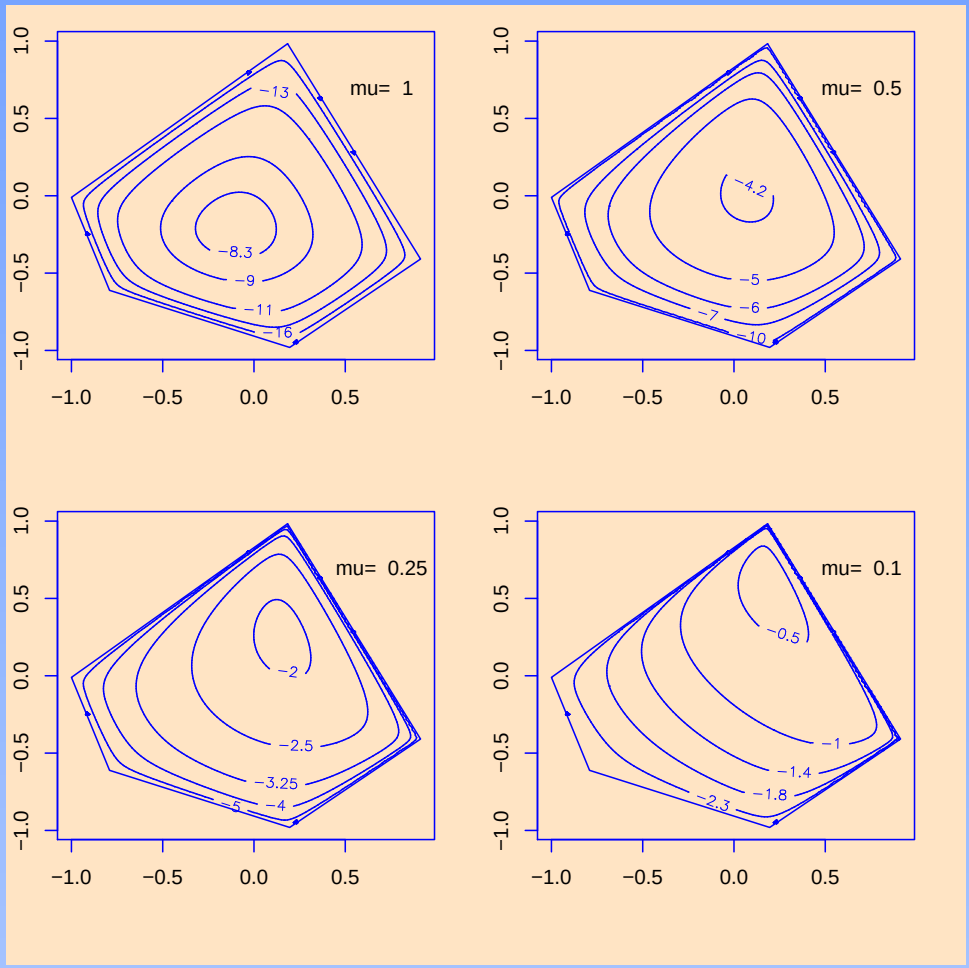
Judiciously, letting $\mu \to 0$ leads us to a solution. The curved central path in the prior figure illustrates the locus of solutions as μ tends to zero.

Contours of the Log Barrier Objective Function



Interior Point Formulation of Quantile Regression

The dual formulation of the quantile regression problem is,

$$\max_a \{y'a \mid X'a = (1-\tau)X'e, \;\; a \in [0,1]^n\},$$

Again we simply replace the inequality constraints with the barrier penalty

$$\max_a \{y'a + \mu \sum (\log a_i + \log(1-a_i)) \mid X'a = (1-\tau)X'1\},$$

and proceed. In the new formulation we can compute Newton steps; Portnoy and Koenker (1997) call this a Frisch-Newton algorithm. With some strategic preprocessing timings are very competitive with least squares even on extremely large problems.

Another Interpretation

"It's about the tortoise ... he's tested positive for steroids."

Some Equivariance Properties of Quantiles Regression

- Scale Equivariance: For any $a > 0$, $\hat{\beta}(\tau; ay, X) = a\hat{\beta}(\tau; y, X)$ and $\hat{\beta}(\tau; -ay, X) = a\hat{\beta}(1-\tau; y, X)$

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- Reparameterization of Design: For any $|A| \neq 0$, $\hat{\beta}(\tau; y, AX) = A^{-1}\hat{\beta}(\tau; yX)$

Equivariance to Monotone Transformations

For any monotone function h, conditional quantile functions $Q_Y(\tau|x)$ are equivariant in the sense that

$$Q_{h(Y)|X}(\tau|x) = h(Q_{Y|X}(\tau|x))$$

In contrast to conditional mean functions for which

$$E(h(Y)|X) \neq h(EY|X)$$

Examples:

$h(y) = \min\{0, y\}$, Powell(1985)

$h(y) = \text{sgn}\{y\}$ Rosenblatt(1957) Manski(1975)

Robustness

- Bounded Influence Function in y for fixed x_i, decent breakdown behavior for fixed design.

- Only the signs of the residuals $\hat{u} = y - X\hat{\beta}(\tau, y, X)$. matter

$$\hat{\beta}(\tau; y, X) = \hat{\beta}(\tau, y + D\hat{u}, X)$$

  for any diagonal matrix D with nonnegative elements. Why? Because the y_i's appear in the optimality (subgradient) condition only through the signs of the residuals.

- Robustness with respect to influential x observations is more challenging, but there are several very interesting proposals. More on this from Ivan Mizera, later.

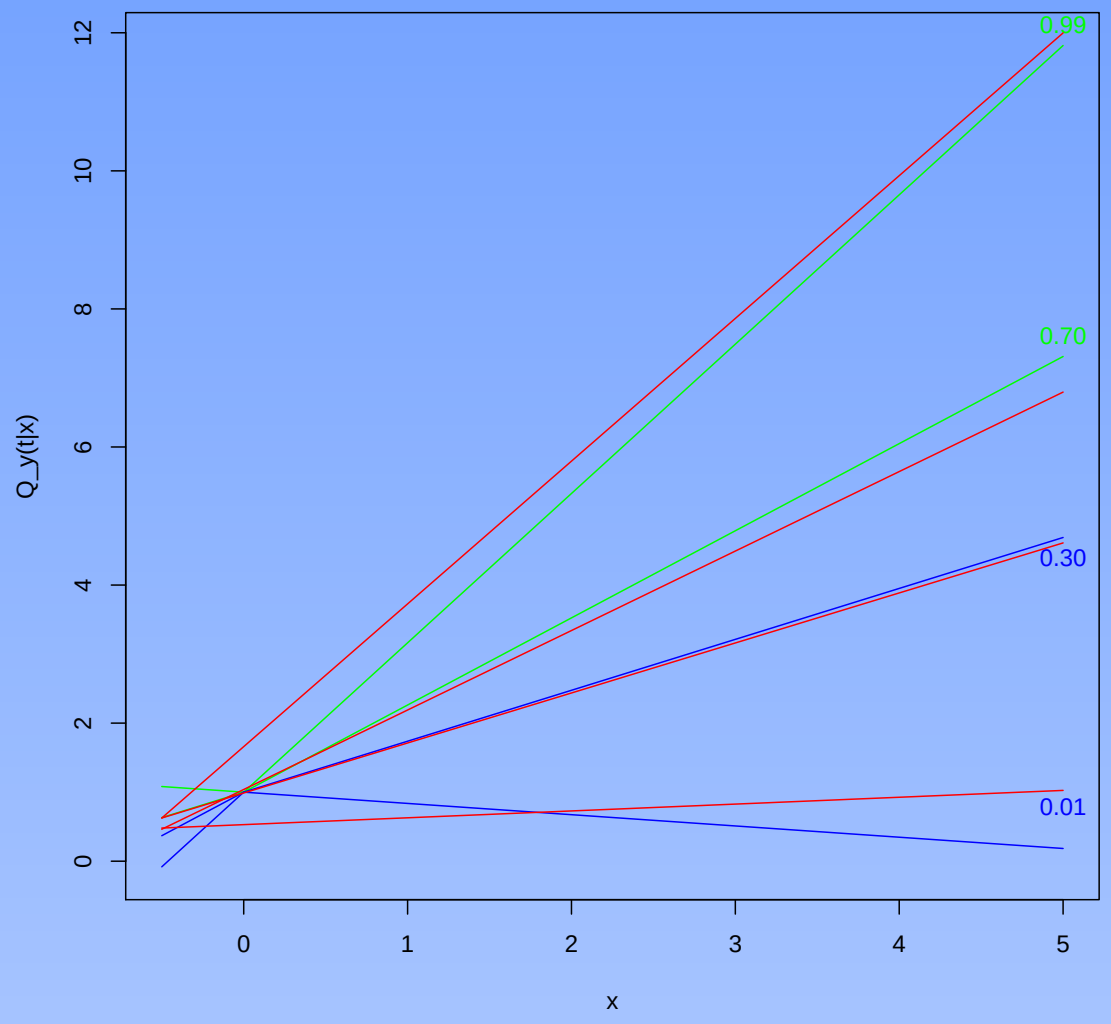
Caution: Quantiles Crossing!

Obviously, $Q_Y(\tau|x)$ should be monotone in τ for all x. However, estimates, $\hat{Q}_Y(\tau|x) = x^\top \hat{\beta}(\tau)$ may violate this principle, especially when x is remote.

Theorem. (Koenker and Bassett (1982)) $\hat{Q}_Y(\tau|\bar{x})$ is monotone.

Remark. Near $\bar{x} = n^{-1} \sum x_i$ one can also expect monotone behavior, but violations of monotonicity can occur in more remote regions of the design space. Misspecification of the functional form of the conditional quantiles can accentuate the likelihood of these violations.

Linear Location-Scale Model I


0.99
0.70
0.30
0.01
Q_y(t|x)
x
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4
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8
10
12
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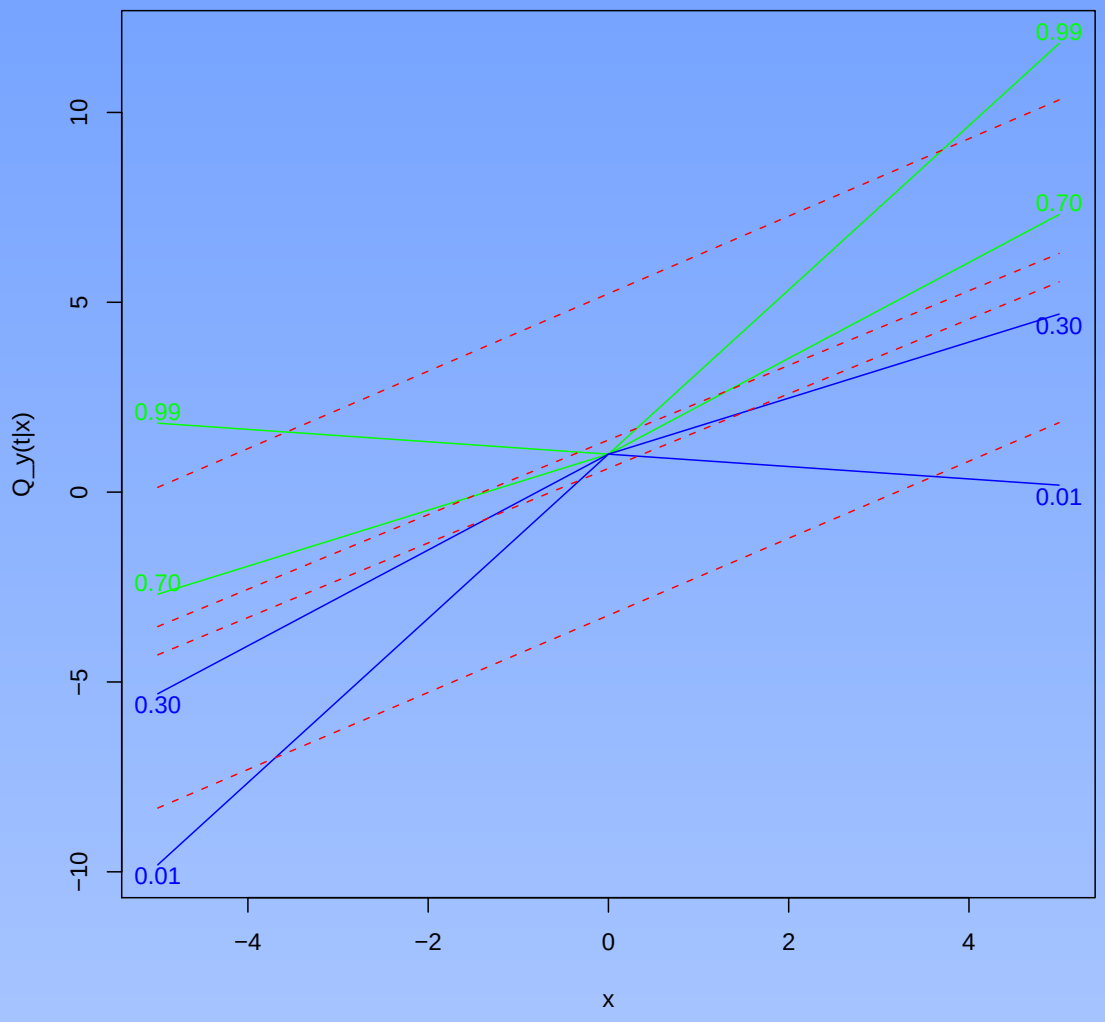
5

Linear Location-Scale Model II


0.99
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Q_y(t|x)
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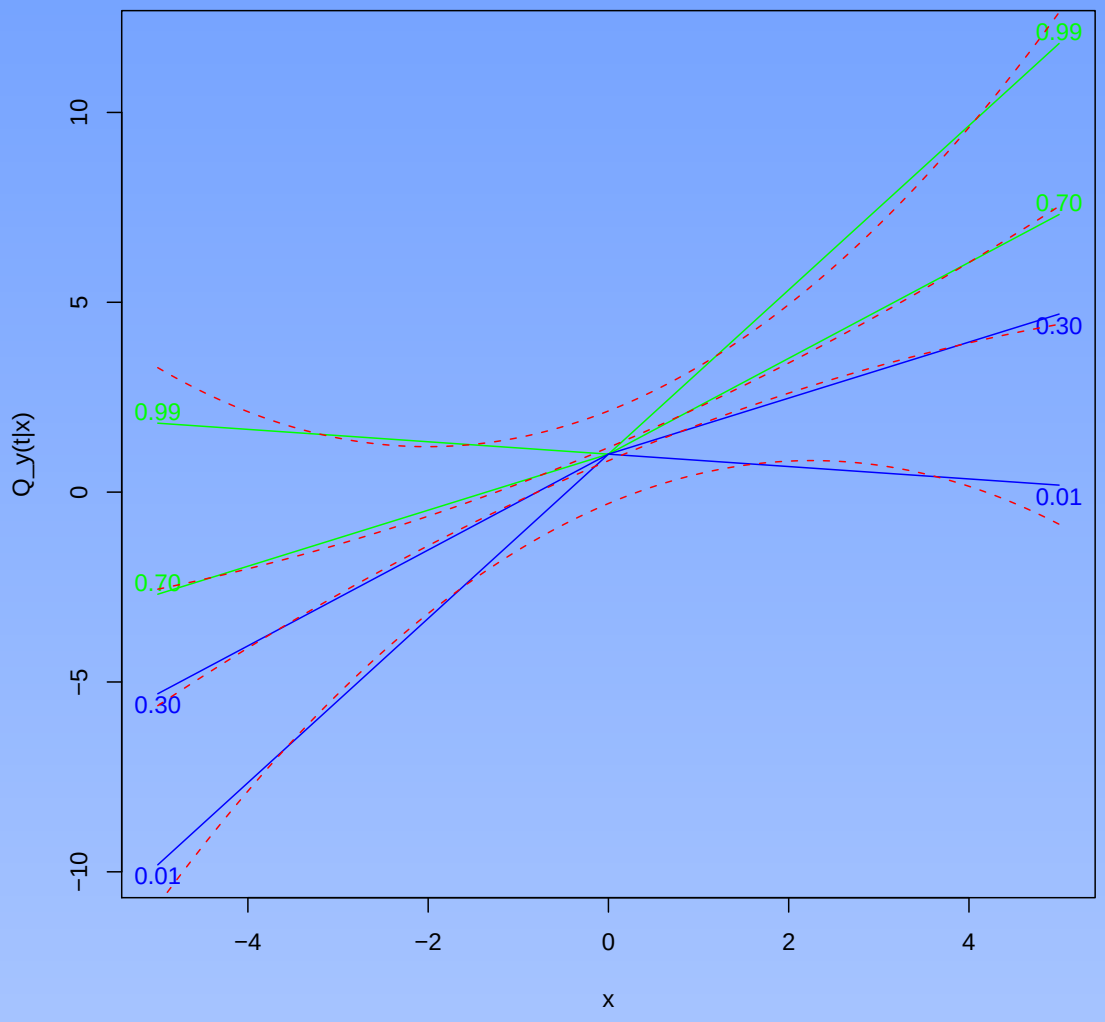
Linear Location-Scale Model III


0.99
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Q_y(t|x)
−4
−2
0
2
4
x

Simulation and Random Coefficients

Recall that:

$$Y \sim F \Rightarrow F(Y) \sim U[0,1]$$
$$U \sim U[0,1] \Rightarrow Y = F^{-1}(U) \sim F.$$

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Thus, we can easily simulate from conditional quantile models by drawing x's from the marginal distribution of X, and then, for $U \sim U[0,1]$ computing

$$Y = Q_{Y|X}(U|x)$$

that is, by replacing τ by the random U. When

$$Q_{Y|X}(\tau|x) = x^\top \beta(\tau)$$

this leads to a useful random coefficient interpretation of the quantile regression model.

Decomposition of Gini's Inequality Coefficient

Gini's Mean Difference: $\gamma = (2\mu)^{-1} E|X_1 - X_2|$, the expected absolute difference in, say, income between two randomly chosen individuals.

Lorenz Curve: $\lambda(t) = \mu^{-1} \int_0^t Q_Y(\tau) d\tau$.

Gini (reformulated): $\gamma = 1 - 2 \int_0^1 \lambda(t) dt$.

Now suppose that we write the conditional quantile function of Y as,

$$Q_h(Y) = (\tau|x) = x^\top \beta(\tau)$$

Decomposition of Conditional Gini's

The conditional Lorenz curve can be decomposed into the additive form:

$$\begin{aligned}\lambda_h(\tau|x) &= \mu_h^{-1}\int_0^t Q_h(Y) = (\tau|x) = x^\top\beta(\tau) \\ &= \mu_h^{-1}\sum_{j=1}^p x_j \int_0^t \beta(\tau)d\tau.\end{aligned}$$

So conditional Gini's can be decomposed into contributions of various covariate effects, and then integrated with respect to the distribution of the covariates to get back to estimates of the unconditional Gini.

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- Simulation of QR models is easy and leads to a random coefficient interpretation of the models,

- Decomposition of the Lorenz curve and Gini coefficient is convenient for additive linear quantile regression models.